SECURE ML GOVERNANCE

& ZERO-DOWNTIME DEPLOYMENT PLATFORM

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TLDR A post-training MLOps platform that **vets** a candidate model, serves it behind a **security gate**, releases it by **canary or shadow traffic**, and **rolls back automatically**. Every prediction is audited and every model is **traceable to its data**.

WHY DEPLOYMENT NEEDS GOVERNANCE

Unvetted releases
A new model version can ship with nobody checking accuracy, determinism or stress behavior.

Exposed inference APIs
Out-of-bounds inputs can crash a model or spike latency.

Big-bang rollouts
Switching 100% of traffic at once turns every bug into an outage.

Silent drift
Live inputs move away from training data and nobody notices.

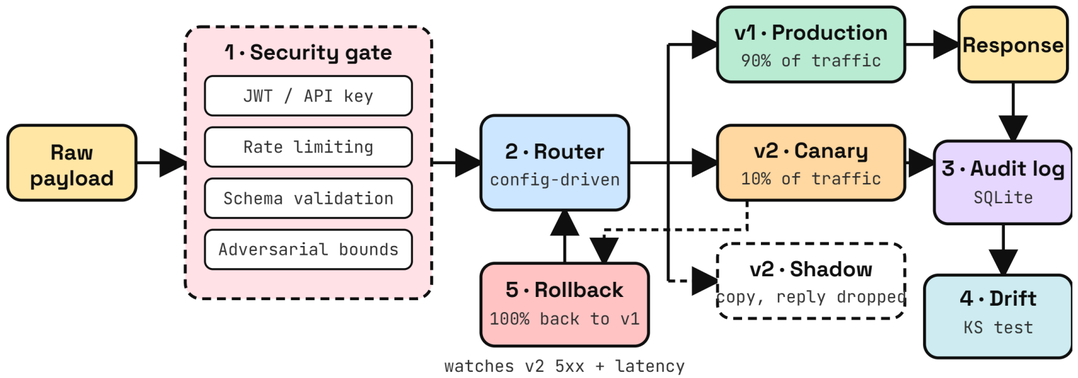
No paper trail
Without lineage and request logs, a prediction can't be traced to its model or data.

OUR ANSWER

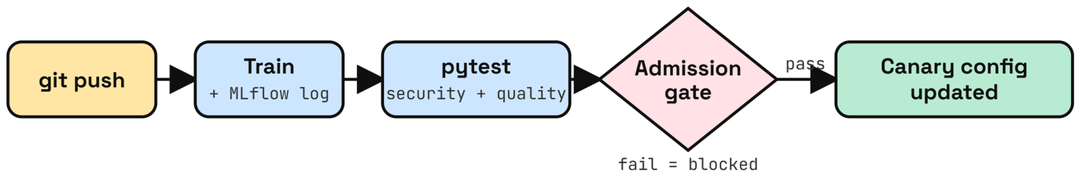
Five guards around one model: security gate, router, audit log, drift monitor and auto-rollback, enforced by a CI/CD admission gate and tracked end to end with MLflow lineage.

Flask · JWT · scikit-learn · SQLite · KS test · PyTest · MLflow · Prometheus · Docker Compose · Gunicorn · GitHub Actions · 28 tests pass

HOW? ONE REQUEST, FIVE GUARDS



GATEKEEPING: CI/CD ADMISSION



The gate checks **artifact loads** · **deterministic output** · **stress handling** · **minimum accuracy** · **demographic-parity gap** (when a reviewed attribute is supplied).

TWO WAYS TO RELEASE

Canary
A weighted split (v1 = 90, v2 = 10) lets **real users** try v2. A breach in error rate or latency **reverts to v1 at once**.

Shadow
A controlled share of live traffic is **copied** to v2. Users only see v1, so latency and accuracy are measured **risk-free**.

MLOPS ADDITIONS

MLflow tracking
Each run logs version, model type, accuracy, **dataset SHA-256** and the quality-gate result. Optional model registry.

Lineage manifest
model_summary.json ties a deployment to its dataset hash, model hash, schema version and MLflow run ID.

Prometheus metrics
Requests, errors, latency, rollback state and traffic weight per model version, ready for **Grafana**.

API ENDPOINTS

`POST /auth/login`	get a JWT
`POST /predict`	routed inference
`GET /health`	models + metadata loaded
`GET /metrics`	traffic, errors, rollback
`GET /metrics/prometheus`	Prometheus text format
`GET /models/metadata`	lineage and hashes
`GET /config`	routing rules
`GET /drift`	drift snapshot
`POST /admin/rollback/reset`	clear rollback